Universidade de Aveiro
Departamento de Electrónica,
Telecomunicações e Informática

Data Visualization module 2026
Introduction

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What is Data Visualization?

- Visualization is focused on how to **visually represent and explore large amounts of data**
- Taking advantage of the **human visual system capacities**
- Providing “**insights**” concerning the phenomenon behind the data

What it **is not**:

- just “pretty pictures”!

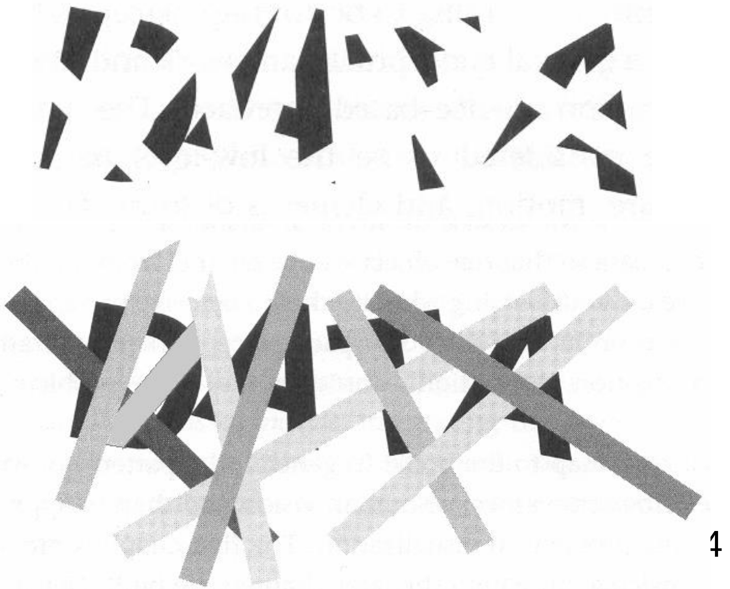
Why and how to represent data visually?

- The human visual system is a most powerful pattern seeker

    “seeing is understanding…”

- We easily see patterns displayed in certain ways

    but not in others …



An exercise in preattentive processing: how many “3”?

69704259347493
58728294954642
44396854634235
6658789376

(Nussbaumer Knaflic, 2015)

69704259**3**4749**3**

58728294954642

44**3**968546**3**42**3**5

6658789**3**76

It is much easier to perform this task!

What if: How many “7”?

C. Nussbaumer Knaflic, Storytelling with Data ,Talks at Google, 2015

[Storytelling with Data | Cole Nussbaumer Knaflic | Talks at Google]

Why represent data visually?

Vis helps in situations where seeing the dataset structure in detail is better than seeing only a brief summary of it.

(Munzner, 2014)

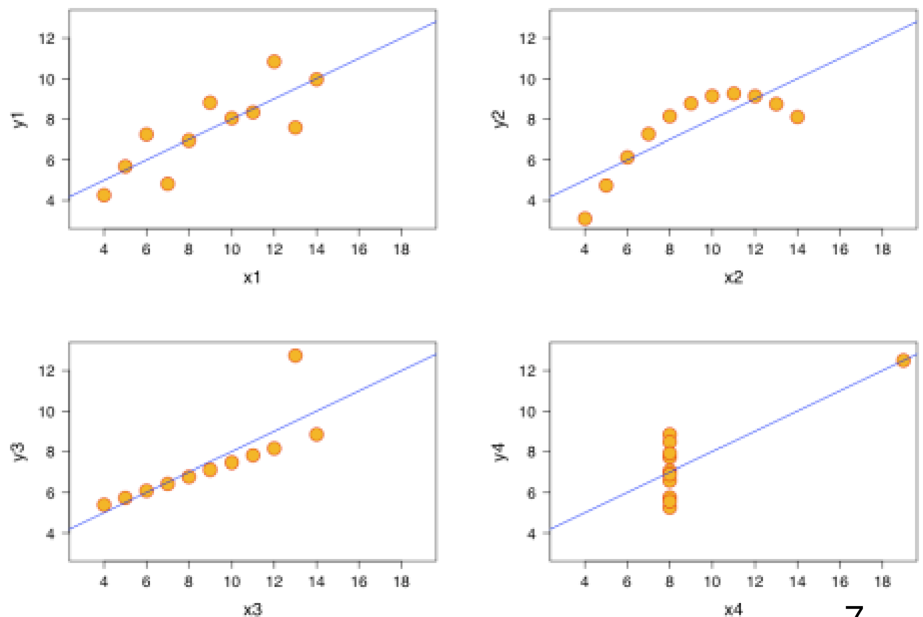
Anscombe's Quartet: Raw Data

	I		II		III		IV	
	x	y	x	y	x	y	x	y
	10.0	8.04	10.0	9.14	10.0	7.46	8.0	6.58
	8.0	6.95	8.0	8.14	8.0	6.77	8.0	5.76
	13.0	7.58	13.0	8.74	13.0	12.74	8.0	7.71
	9.0	8.81	9.0	8.77	9.0	7.11	8.0	8.84
	11.0	8.33	11.0	9.26	11.0	7.81	8.0	8.47
	14.0	9.96	14.0	8.10	14.0	8.84	8.0	7.04
	6.0	7.24	6.0	6.13	6.0	6.08	8.0	5.25
	4.0	4.26	4.0	3.10	4.0	5.39	19.0	12.50
	12.0	10.84	12.0	9.13	12.0	8.15	8.0	5.56
	7.0	4.82	7.0	7.26	7.0	6.42	8.0	7.91
	5.0	5.68	5.0	4.74	5.0	5.73	8.0	6.89
mean	9.0	7.5	9.0	7.5	9.0	7.5	9.0	7.5
var.	10.0	3.75	10.0	3.75	10.0	3.75	10.0	3.75
corr.	0.816		0.816		0.816		0.816	

[Anscombe's quartet - Wikipedia]

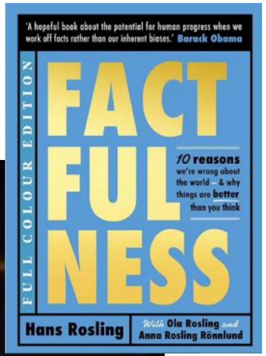
Anscombe quartet: data sets with the same simple statistical properties

(Tufte, 1983)



Example of Presentation based on a simple Visualization: World health by Hans Rosling

200 years of health/income – **120 000 values in 4 min**



[Hans Rosling's 200 Countries, 200 Years, 4 Minutes - The Joy of Stats – BBC](#)

[Factfulness Illustrated: Ten Reasons We're Wrong About the World - Why ... - Hans Rosling, Ola Rosling, Anna Rosling Rönnlund - Google Books](#)

This module:

- Objective, bibliography and tools
- Introduction to Data Visualization and its application
- Data and phenomena: types and pre-processing
- Human-in-the-loop process: perceptual and cognitive aspects
- Visualization of tabular and field data:

    1D and 2D Data – main methods; exercises

    3D and nD data - main methods; exercises
- Visualization of other types of data

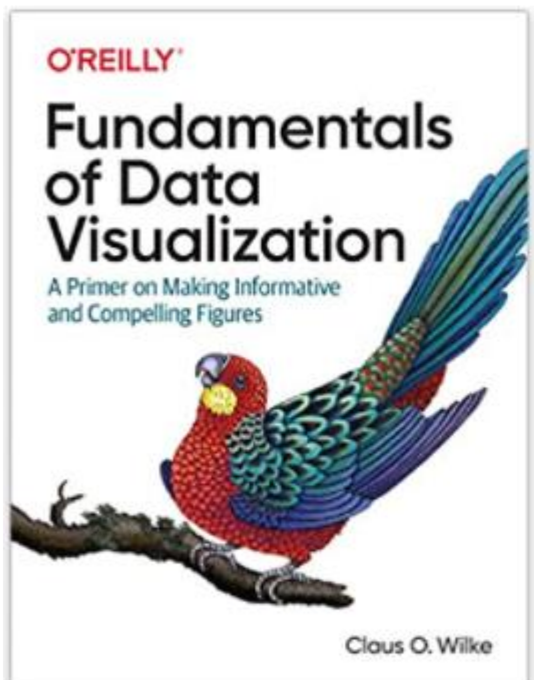
    georeferenced, networks, hierarchical data, etc.
- Effective Visualization

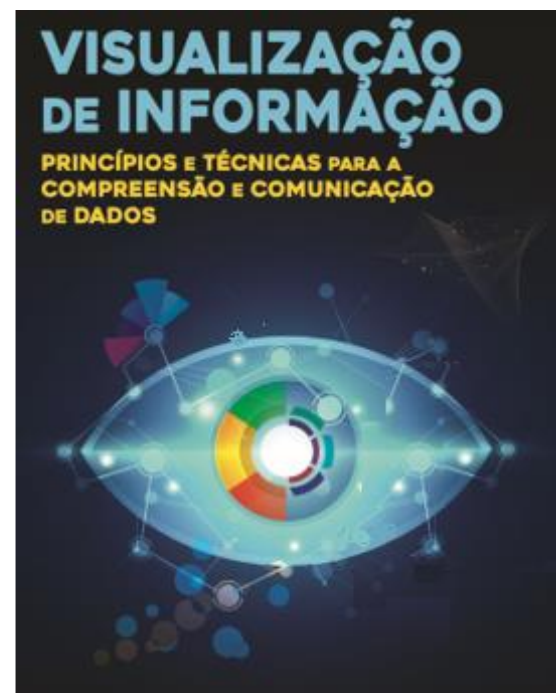
Bibliography

- Camões, J., *Data at Work : Best practices for creating effective charts and information graphics in Microsoft Excel*, Pearson Education, 2016

- Wilke, C., *Fundamentals of Data Visualization: A Primer on Making Informative and Compelling Figures*, O'Reilly, 2019

- Gama, S., Gonçalves, D., Sousa Santos, B., Moreira, J., *Visualização de Informação: Princípios e Técnicas para a Compreensão e Comunicação de Dados*, FCA, 2025



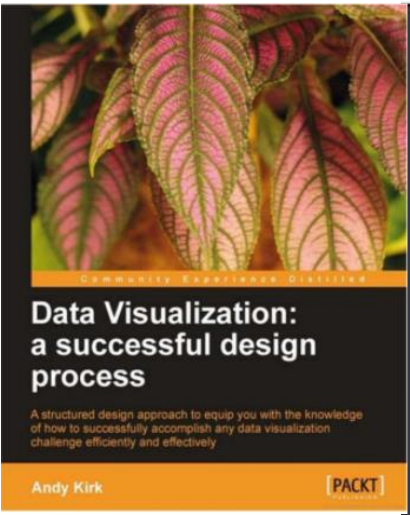


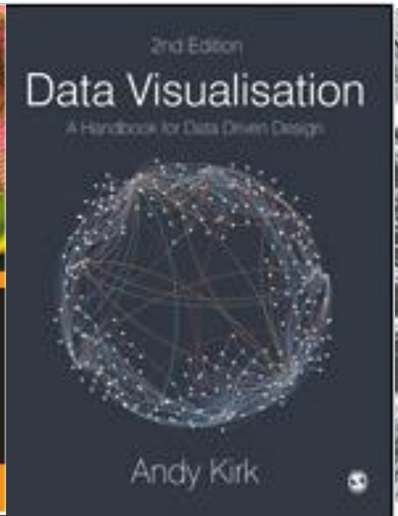
More advanced bibliography

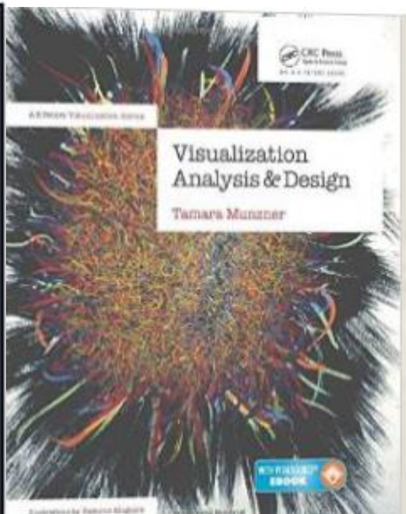
- Kirk, A., Data Visualisation A Handbook for Data Driven Design, 2nd. Ed., Sage, 2019
- Kirk, A., *Data Visualization: A successful design process*, Pack Publishing, 2012
- Munzner, T., *Visualization Analysis and Design*, CRC Press, 2014
- Telea, A., *Data Visualization: Principles and Practice*, CRC Press, 2015
- Ware, C., Information Visualization, Perception for Design, 3rd ed., Morgan Kaufman, 2012

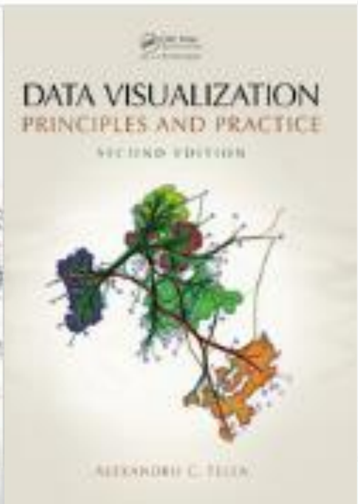
Explore yet other books available at the playlist:

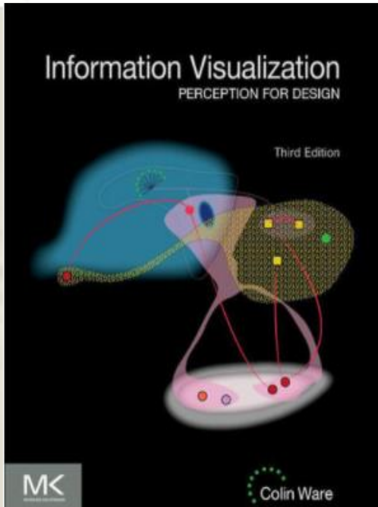
https://learning.oreilly.com/playlists/74bfec5e-4346-48ff-82b4-657fda6922b6





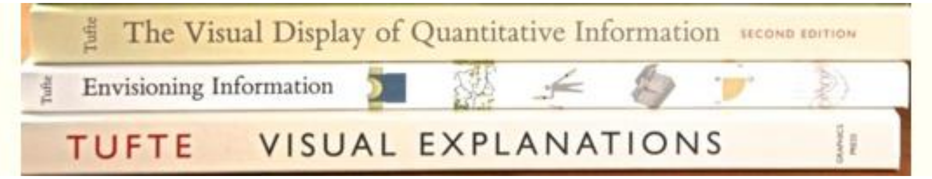






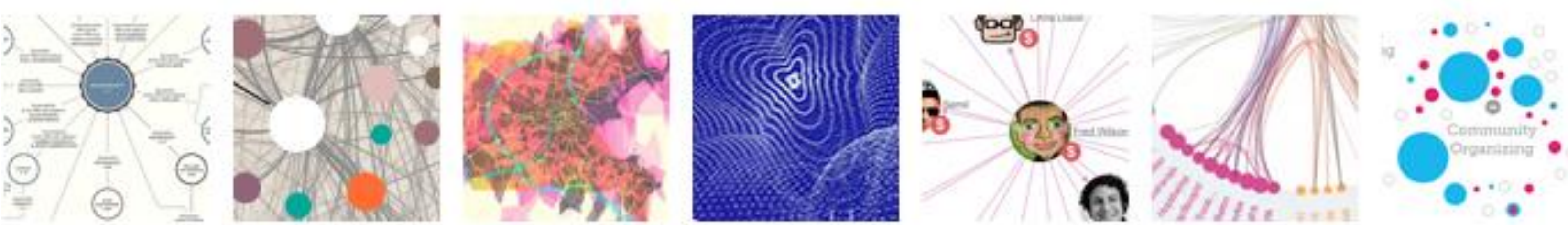
Interesting links

- Everything by Jorge Camoes
- Eagereyes.org
- Perceptual Edge
- The Work of Edward Tufte & Graphics Press



Interesting links

- Multiple Views: Visualization Research Explained
- Home - Seeing Data
- FlowingData
- VisualComplexity.com. 10 Years / 1000 Projects | by Manuel Lima



Moodle

Slides

- Introduction to the module
- Introduction to Visualization
- Data
- Creating a Visualization
- Effective Visualization

Bibliography

- Data - Basic concepts
- Creating a Visualization
- Camões, J., Data at Work : Best practices for creating effective charts and information graphics in Microsoft Excel, Pearson Education, 2016
- Wilke, C., Fundamentals of Data Visualization: A Primer on Making Informative and Compelling Figures, O'Reilly, 2019
- Healy, K., Data Visualization a Practical Introduction, Princeton, 2019
- Kirk, A., Data Visualization: A successful design process, Pack Publishing, 2012
- Spence, R., Design for Interaction (2nd Edition), Pearson, 2007
- More Advanced Bibliography- playlist

Sessions
(subject to minor adjustments)

1 - Introduction to the module. Data Visualization – overview and applications

2 -Effective Visualization; application examples and critical analysis of visualizations

3 – Visualization model. Data - different types, phenomena they represent and pre-processing. Visualization Literacy Quiz

4 - Creating a Visualization – questions guiding the design; visual encoding
Visualization of 1D and 2D, 3D tabular data – main techniques; simple exercises

5 – Application exercises, quizzes and critical analysis of visualizations

6 - Visualization of multivariate tabular data and fields – main techniques; examples

7 – Application exercises, critical analysis of visualizations; quiz

8 - Visualization of relation and hierarchical data – main techniques; quiz

9 - Test

Assessment Module 2

Select assessment mode asap!

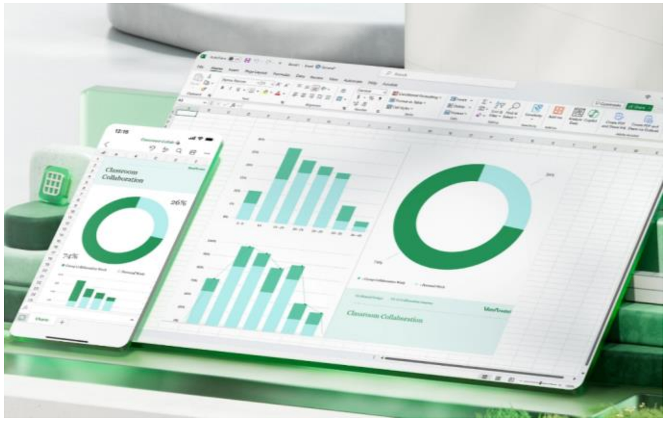
Check if this is your CT II module!!

During the semester:
Test – 100%

Alternative at the end of semester:
Exam – 100%, June

Test – last session of the module – April,23rd

Practical activities



Quizzes, exercises and critical analysis

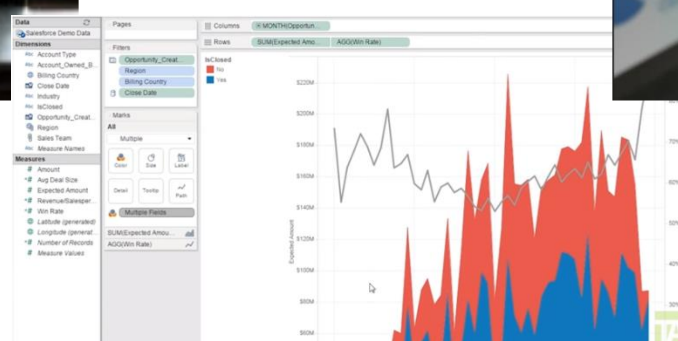
Microsoft-365/excel

Matplotlib

Next lecture bring your own laptop!

Questions regarding the module?

Visualization S/W



Which Visualization Tool?
It depends …

- Exploratory (discover pattern, multiple views)

  or Explanatory (View of the data presenting discovered highlights)?
- Type of data (Maps, Charts, Data,…)
- Developer or non-developer?
- Scientific or information Visualization (2D,3D, structured or not?)
- Interactive or Static?
- Web or local?
- Easy to use or Flexible?
- Protection of data?…

Visualization Tools – Some possible choices

- If you want to produce a few simple charts for a report or paper: Excel, …
- If you want to produce some charts and have some programming skills: MATLAB, R, python libraries, …
- If you use some statistics/analytics S/W: Statistica, SPSS, …
- If you are in a large company: Tableau or Qlickview may be adequate (very powerful and expensive business intelligence S/W)
- If you just want to make a few simple charts for your web page and have programming skills: google charts
- If you want to develop an interactive visualization Web application to visually explore data: D3.js

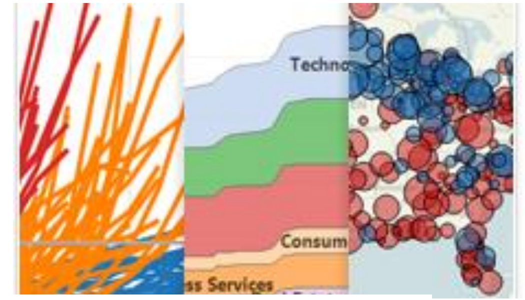
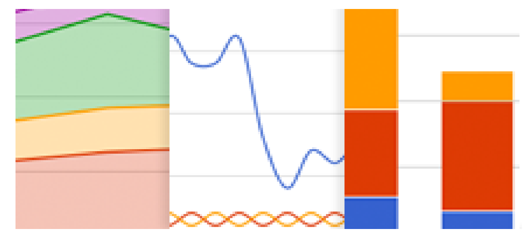


Tableau Public

A desktop application to build and post interactive graphs, dashboards, maps and tables to the web.



Google Chart Tools

A collection of simple to use, customizable and free to use interactive charts and data tools.

D3.js

An small, flexible and efficient library to create and manipulate interactive documents based on data.

Visualization Tools – Possible choices

Datavisualization.ch Selected Tools – Filters available

- Maps
- Charts
- Data
- Color

- Developer
- Non-developer

We will use

- Excel
- Flourish
- Python + Matplotlib

(but you may use also other S/W if you master it)